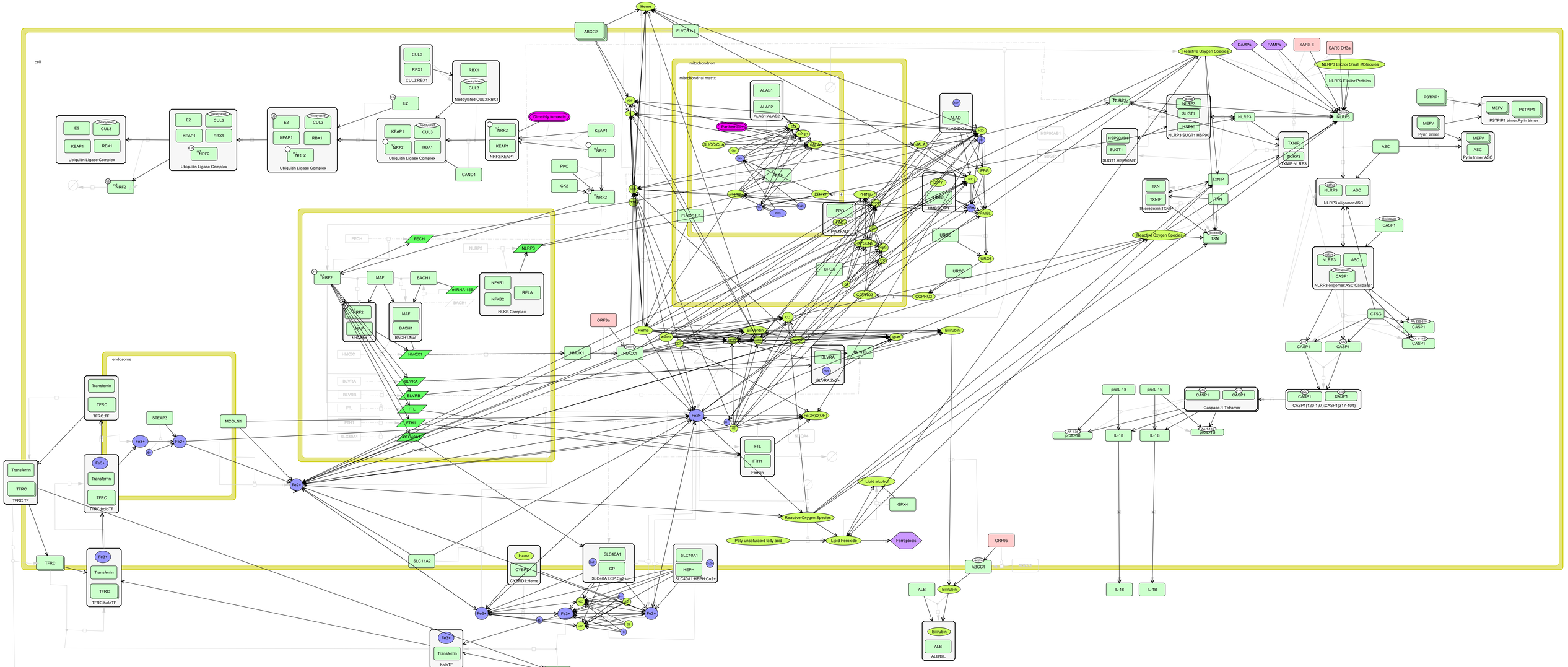

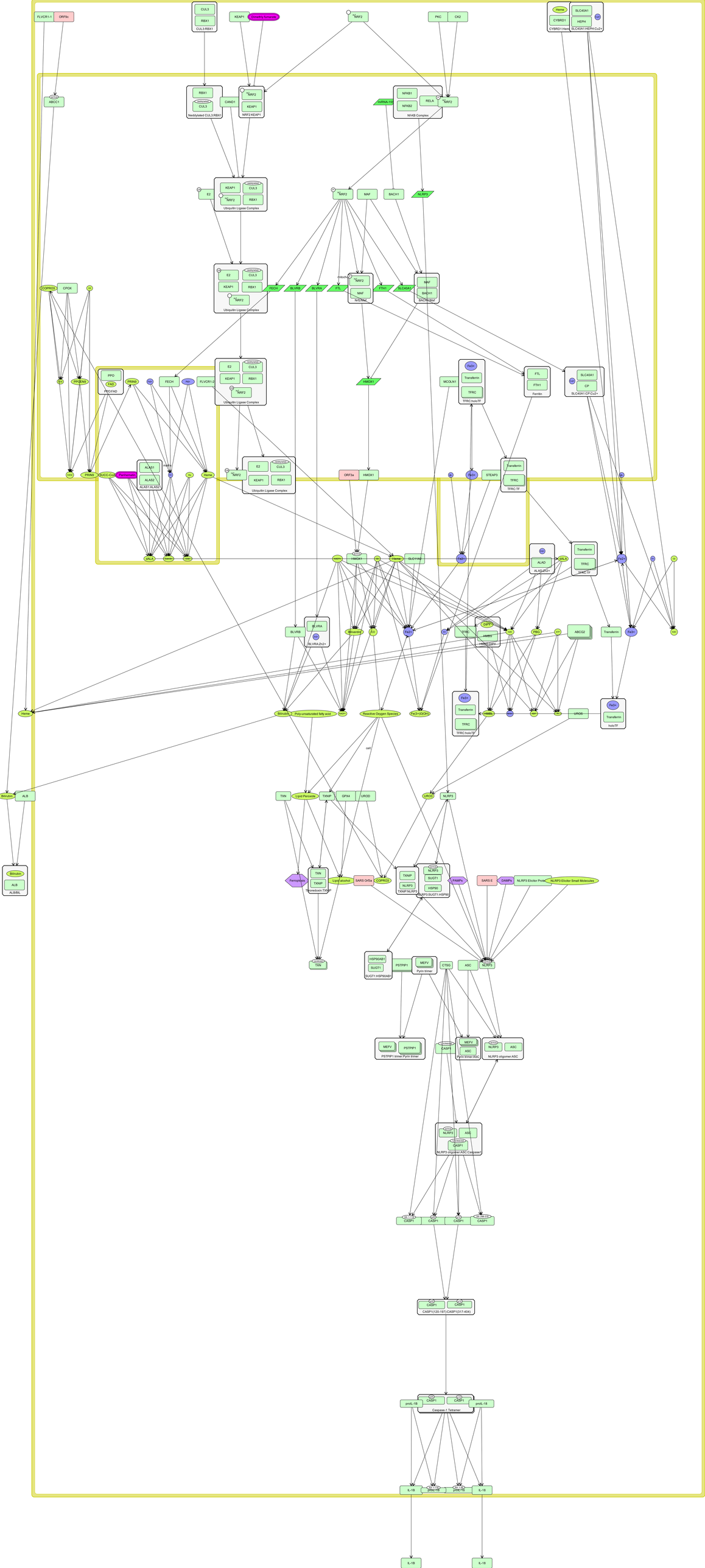

cell
CUL3
RBX1
CUL3:RBX1
neddylated
Neddylated CUL3:RBX1
Ub
E2
KEAP1
rs1 NRF2
Ubiquitin Ligase Complex
NRF2:KEAP1
CAND1
Dimethly fumarate
PKC
CK2
P
ABCG2
Heme
FLVCR1-1
FLVCR1-2
ADP
Pi
H2O
ATP
mitochondrion
mitochondrial matrix
ALAS1
ALAS2
ALAS1:ALAS2
Panhematin
SUCC-CoA
Gly
CO2
CoA-SH
dALA
H+
FECH
PRIN9
Fe2+
Pb2+
PPO
FAD
PPO:FAD
PPGEN9
H2O2
O2
CPOX
COPRO3
Zn2+
ALAD
ALAD:Zn2+
PBG
DIPY
HMBS
HMBS:DIPY
NH4+
HMBL
UROS
URO3
UROD
HSP90AB1
SUGT1
SUGT1:HSP90AB1
NLRP3
active
HSP90
NLRP3:SUGT1:HSP90
Reactive Oxygen Species
DAMPs
PAMPs
SARS E
SARS Orf3a
NLRP3 Elicitor Small Molecules
NLRP3 Elicitor Proteins
TXNIP
TXNIP:NLRP3
TXN
Thioredoxin:TXNIP
oxidized
PSTPIP1
MEFV
PSTPIP1 trimer:Pyrin trimer
Pyrin trimer
ASC
Pyrin trimer:ASC
NLRP3 oligomer:ASC
Uncleaved
CASP1
NLRP3 oligomer:ASC:Caspase1
CTSG
AA 298-316
AA 1-119
p10
p20
CASP1(120-197):CASP1(317-404)
Caspase-1 Tetramer
proIL-18
proIL-1B
AA 1-36
AA 1-116
IL-18
IL-1B
MAF
BACH1
miRNA-155
Nrf2/Maf
BACH1/Maf
NFKB1
NFKB2
RELA
Nf-KB Complex
HMOX1
BLVRA
BLVRB
FTL
FTH1
SLC40A1
nucleus
ORF3a
CO
Biliverdin
Bilirubin
NADPH
NADP+
BLVRA:Zn2+
Fe(3+)O(OH)
NCOA4
Ferritin
Lipid alcohol
GPX4
Poly-unsaturated fatty acid
Lipid Peroxide
Ferroptosis
ORF9c
ABCC1
ALB
ALB/BIL
endosome
Transferrin
TFRC
TFRC:TF
Fe3+
TFRC:holoTF
STEAP3
e-
MCOLN1
SLC11A2
CYBRD1
CYBRD1:Heme
Cu2+
CP
SLC40A1:CP:Cu2+
HEPH
SLC40A1:HEPH:Cu2+
holoTF

FLVCR1-1
ORF9c
CUL3
RBX1
CUL3:RBX1
KEAP1
Dimethyl fumarate
rs1 NRF2
PKC
CK2
Heme
CYBRD1
CYBRD1:Heme
SLC40A1
HEPH
Cu2+
SLC40A1:HEPH:Cu2+
active
ABCC1
RBX1
neddylated
CUL3
Neddylated CUL3:RBX1
CAND1
rs1 NRF2
KEAP1
NRF2:KEAP1
NFKB1
NFKB2
RELA
P
rs1 NRF2
Nf-KB Complex
miRNA-155
Ub
E2
KEAP1
neddylated
CUL3
rs1 NRF2
RBX1
Ubiquitin Ligase Complex
P
rs1 NRF2
MAF
BACH1
NLRP3
Ub
E2
neddylated
CUL3
KEAP1
RBX1
rs1 NRF2
Ubiquitin Ligase Complex
FECH
BLVRB
BLVRA
FTL
mitochondrion
P
rs1 NRF2
MAF
Nrf2/Maf
FTH1
SLC40A1
MAF
BACH1
BACH1/Maf
COPRO3
CPOX
O2
PPO
FAD
PPO:FAD
PRIN9
Fe2+
FECH
Pb2+
FLVCR1-2
E2
neddylated
CUL3
KEAP1
RBX1
Ub
rs1 NRF2
Ubiquitin Ligase Complex
HMOX1
MCOLN1
Fe3+
Transferrin
TFRC
TFRC:holoTF
FTL
FTH1
Ferritin
SLC40A1
Cu2+
CP
SLC40A1:CP:Cu2+
CO2
PPGEN9
H2O2
PRIN9
SUCC-CoA
Panhematin
ALAS1
ALAS2
ALAS1:ALAS2
matrix
H+
Gly
Heme
Ub
rs1 NRF2
E2
neddylated
CUL3
KEAP1
RBX1
Ubiquitin Ligase Complex
ORF3a
HMOX1
e-
Fe3+
STEAP3
Transferrin
TFRC
TFRC:TF
e-
dALA
CoA-SH
CO2
NADPH
active
HMOX1
O2
Heme
SLC11A2
Fe2+
Zn2+
ALAD
ALAD:Zn2+
dALA
Transferrin
TFRC
TFRC:TF
Fe2+
H+
O2
BLVRB
BLVRA
Zn2+
BLVRA:Zn2+
Biliverdin
CO
Fe2+
H+
TFRC
DIPY
HMBS
HMBS:DIPY
H2O
PBG
ATP
ABCG2
Transferrin
Fe3+
H2O
Heme
Bilirubin
Poly-unsaturated fatty acid
NADP+
Reactive Oxygen Species
Fe(3+)O(OH)
Fe3+
Transferrin
TFRC
TFRC:holoTF
HMBL
NH4+
ADP
Pi
UROS
Fe3+
Transferrin
holoTF
cell
Bilirubin
ALB
TXN
Lipid Peroxide
TXNIP
GPX4
UROD
URO3
NLRP3
Bilirubin
ALB
ALB/BIL
Ferroptosis
TXN
TXNIP
Thioredoxin:TXNIP
Lipid alcohol
SARS Orf3a
COPRO3
TXNIP
NLRP3
TXNIP:NLRP3
active
NLRP3
SUGT1
HSP90
NLRP3:SUGT1:HSP90
PAMPs
SARS E
DAMPs
NLRP3 Elicitor Proteins
NLRP3 Elicitor Small Molecules
oxidized
TXN
HSP90AB1
SUGT1
SUGT1:HSP90AB1
PSTPIP1
MEFV
Pyrin trimer
CTSG
ASC
active
NLRP3
MEFV
PSTPIP1
PSTPIP1 trimer:Pyrin trimer
Uncleaved
CASP1
MEFV
ASC
Pyrin trimer:ASC
active
NLRP3
ASC
NLRP3 oligomer:ASC
active
NLRP3
ASC
Uncleaved
CASP1
NLRP3 oligomer:ASC:Caspase1
AA 1-119
CASP1
p20
CASP1
p10
CASP1
AA 298-316
CASP1
p20
CASP1
p10
CASP1
CASP1(120-197):CASP1(317-404)
proIL-1B
p20
CASP1
p10
CASP1
Caspase-1 Tetramer
proIL-18
IL-1B
AA 1-116
proIL-1B
AA 1-36
proIL-18
IL-18
IL-1B
IL-18